TUGAS 05

Praktikum Bab 5

Matakuliah Workshop DevOps

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Oktober 2026

Praktikum Devsecops 5

Kotak Perintah A – Menyiapkan Direktori Bab 5

```
toimul@DESKTOP-2OIAP1F:~$ mkdir -p ~/docker-lab/bab-5/{init,ba
ckup,pgadmin,scripts}
toimul@DESKTOP-2OIAP1F:~$ cd ~docker-lab/bab-5
-bash: cd: ~docker-lab/bab-5: No such file or directory
toimul@DESKTOP-2OIAP1F:~$ cd ~/docker-lab/bab-5
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ ls -R
.:
backup  init  pgadmin  scripts

./backup:

./init:

./pgadmin:

./scripts:
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$
```

Script 1 -file .env.example

```
./scripts:
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ nano .env.example
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$
```

```
toimul@DESKTOP-2OIAP1F: ~
  GNU nano 8.7.1                .env.example
POSTGRES_DB=labdb
POSTGRES_USER=labuser
POSTGRES_PASSWORD=labpass123
PGADMIN_DEFAULT_EMAIL=admin@example.local
PGADMIN_DEFAULT_PASSWORD=admin123
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ cp .env.example .en
v
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ chmod 600 .env
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$
```

Script 2- file .gitignore

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ nano .gitignore
```

```
toimul@DESKTOP-2OIAP1F: ~
 GNU nano 8.7.1                .gitignore *
.env
backup/*.dump
backup/*.sha256
backup/*.log
```

Script 3- file compose.yaml

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ nano compose.yaml
```

```
toimul@DESKTOP-2OIAP1F: ~
 GNU nano 8.7.1                compose.yaml
services:
  postgres-db:
    image: postgres:16-alpine
    environment:
      POSTGRES_DB: ${POSTGRES_DB}
      POSTGRES_USER: ${POSTGRES_USER}
      POSTGRES_PASSWORD: ${POSTGRES_PASSWORD}
    ports:
      - "127.0.0.1:5432:5432"
    volumes:
      - pg-data:/var/lib/postgresql/data
      - ./init:/docker-entrypoint-initdb.d:ro
    networks:
      - data-net
    healthcheck:
      test:
        - CMD-SHELL
        - "pg_isready -U $${POSTGRES_USER} -d $${POSTGRES_DB}"
      interval: 5s
      timeout: 5s
      retries: 10
      start_period: 10s
    restart: unless-stopped

  pgadmin:
    image: dpage/pgadmin4:latest
    environment:
```

Script 4- file init/01-schema.sql

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ nano init/01-schema
.sql
```

toimul@DESKTOP-2OIAP1F: ~

```
  GNU nano 8.7.1          init/01-schema.sql *
CREATE TABLE IF NOT EXISTS students (
    id BIGSERIAL PRIMARY KEY,
    nrp VARCHAR(20) UNIQUE NOT NULL,
    name TEXT NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

INSERT INTO students (nrp, name)
VALUES
    ('31230001', 'Mahasiswa Satu'),
    ('31230002', 'Mahasiswa Dua')
ON CONFLICT (nrp) DO NOTHING;

CREATE INDEX IF NOT EXISTS idx_students_name
    ON students (name);
```

Script 5- file pgadmin/servers.json

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ nano pgadmin/server
s.json
```

```
  GNU nano 8.7.1          pgadmin/servers.json *
{
  "Servers": {
    "1": {
      "Name": "PostgreSQL Bab 5",
      "Group": "Laboratorium DevSecOps",
      "Host": "postgres-db",
      "Port": 5432,
      "MaintenanceDB": "labdb",
      "Username": "labuser",
      "SSLMode": "prefer"
    }
  }
}
```

Script 6- file script/backup.sh

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ nano scripts/backup
:sh
```

```
  GNU nano 8.7.1              scripts/backup:sh *

docker compose exec -T postgres-db \
    pg_dump \
    --username "$POSTGRES_USER" \
    --dbname "$POSTGRES_DB" \
    --format custom \
    --no-owner \
    --no-privileges \
    > "$dump_file"

test -s "$dump_file"
sha256sum "$dump_file" > "$checksum_file"

echo "[PASS] Backup: $dump_file"
echo "[PASS] Checksum: $checksum_file"
```

Script 7- file scripts/restore-test.sh

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ nano scripts/restor
e-test.sh
```

```
  GNU nano 8.7.1          scripts/restore-test.sh *
#!/usr/bin/env bash
set -Eeuo pipefail

cd "$(dirname "$0")/.."

if [[ ! -f .env ]]; then
    echo "[FAIL] File .env tidak ditemukan." >&2
    exit 1
fi

set -a
source .env
set +a
```

```
dump_file="${1:-}"
restore_database="${POSTGRES_DB}_restore_test"

if [[ -z "$dump_file" || ! -s "$dump_file" ]]; then
    echo "Penggunaan: $0 backup/nama-file.dump" >&2
    exit 1
fi

if [[ -f "${dump_file}.sha256" ]]; then
    sha256sum --check "${dump_file}.sha256"
fi

docker compose exec -T postgres-db \
    dropdb --username "$POSTGRES_USER" --if-exists "$restore_>
```

Perintah B – Permission dan Validasi File

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ chmod +x scripts/ba
ckup.sh scripts/restore-test.sh
chmod: cannot access 'scripts/backup.sh': No such file or dire
ctory
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ find . -maxdepth 3
-type f | sort
./.env
./.env.example
./.gitignore
./compose.yaml
./init/01-schema.sql
./pgadmin/servers.json
./scripts/backup:sh
./scripts/restore-test.sh
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose conf
ig
name: bab-5
services:
  pgadmin:
    depends_on:
      postgres-db:
        condition: service_healthy
        required: true
    environment:
      PGADMIN_DEFAULT_EMAIL: admin@example.local
      PGADMIN_DEFAULT_PASSWORD: admin123
    image: dpage/pgadmin4:latest
    networks:
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose conf
ig --services
postgres-db
pgadmin
```

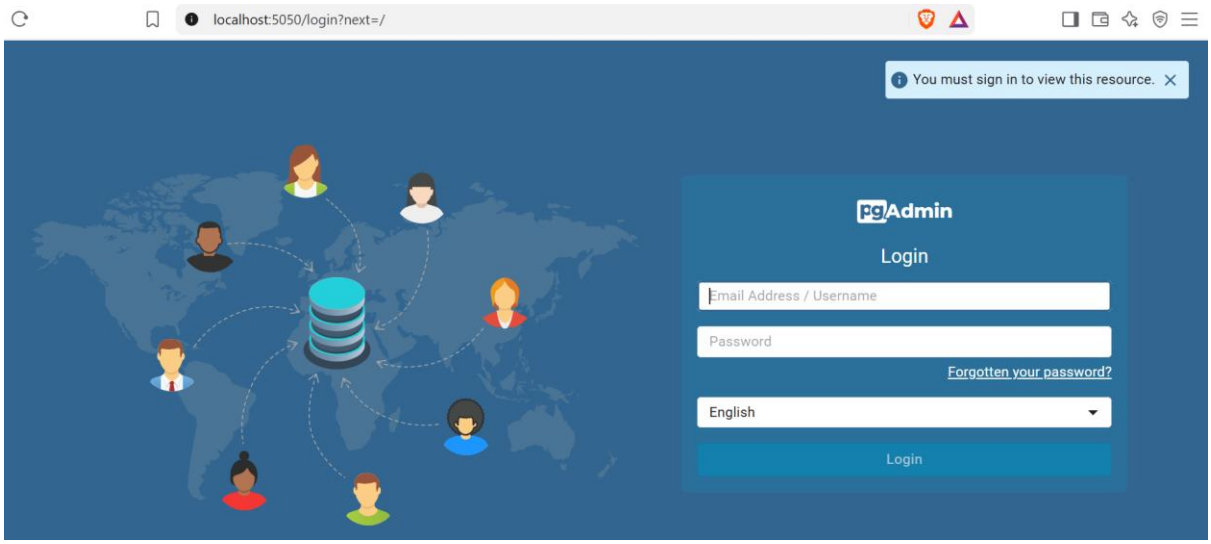
Perintah C – Menjalankan Stack

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose up -
d
[+] up 22/22
 ✔Image dpage/pgadmin4:latest    Pulled                  76.6s
 ✔Network bab-5_data-net         Created                  0.1s
 ✔Volume bab-5_pgadmin-data      Created                  0.0s
 ✔Volume bab-5_pg-data           Created                  0.0s
 ✔Container bab-5-postgres-db-1  Healthy                  8.2s
 ✔Container bab-5-pgadmin-1      Started                  7.3s
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$
```

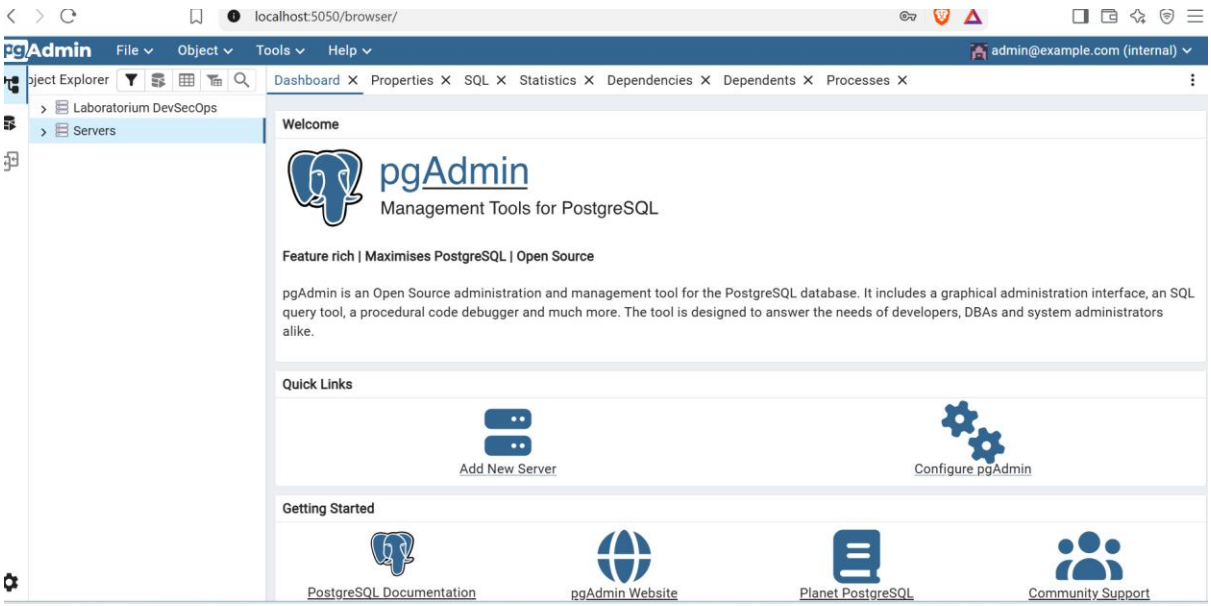
```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose ps
NAME                   IMAGE                      COMMAND
         SERVICE       CREATED          STATUS
         PORTS
bab-5-pgadmin-1        dpage/pgadmin4:latest      "/entrypoint.sh"
         pgadmin       40 seconds ago   Restarting (1) 9 secon
ds ago
bab-5-postgres-db-1    postgres:16-alpine         "docker-entrypoi
nt.s…"   postgres-db   41 seconds ago   Up 39 seconds (healthy
)        127.0.0.1:5432->5432/tcp
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose logs
 --tail 100
pgadmin-1       | email config is {'CHECK_EMAIL_DELIVERABILITY'
: False, 'ALLOW_SPECIAL_EMAIL_DOMAINS': [], 'GLOBALLY_DELIVERA
BLE': True}
pgadmin-1       | 'admin@example.local' does not appear to be a
 valid email address. Please reset the PGADMIN_DEFAULT_EMAIL e
nvironment variable and try again.
pgadmin-1       | Validation output: The part after the @-sign
is a special-use or reserved name that cannot be used with ema
il.
```

Catatan: Sebelumnya localhost:5050 tidak dapat diakses, lalu dilakukan perubahan pada email di .env. .local diganti .com baru bisa diakses.



Setelah login menggunakan email & pass default dari file .env



Perintah D – Menguji Schema dan Data Awal

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db psql -U labuser -d labdb -c "\dt"
          List of relations
 Schema |   Name   | Type  |  Owner
--------+----------+-------+---------
 public | students | table | labuser
(1 row)
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db psql -U labuser -d labdb -c "SELECT id, nrp, n
ame, created_at FROM students ORDER BY id;"
 id |   nrp    |      name      |          created_at
----+----------+----------------+----------------------------
--
  1 | 31230001 | Mahasiswa Satu | 2026-10-10 07:39:11.747413+0
0
  2 | 31230002 | Mahasiswa Dua  | 2026-10-10 07:39:11.747413+0
0
(2 rows)
```

Perintah E – Menguji Persistensi Volume

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db psql -U labuser -d labdb -c "INSERT INTO stude
nts(nrp, name) VALUE ('31230001', 'Mahasiswa')"
ERROR:  syntax error at or near "VALUE"
LINE 1: INSERT INTO students(nrp, name) VALUE ('31230001', 'Ma
hasisw...
                                             ^
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose down
[+] down 3/3
 ✔Container bab-5-pgadmin-1     Removed                  3.1s
 ✔Container bab-5-postgres-db-1 Removed                  0.7s
 ✔Network bab-5_data-net        Removed                  0.3s
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose up -
d
[+] up 3/3
 ✔Network bab-5_data-net        Created                  0.1s
 ✔Container bab-5-postgres-db-1 Healthy                  6.9s
 ✔Container bab-5-pgadmin-1     Started                  7.3s
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db psql -U labuser -d labdb -c "SELECT nrp, name
FROM students ORDER BY nrp"
   nrp    |      name
----------+----------------
 31230001 | Mahasiswa Satu
 31230002 | Mahasiswa Dua
(2 rows)
```

Perintah F – Backup

Pada perintah ini ada perbaikan dan penyesuaian kerana adanya bug atau error

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ nano scripts/backup
.sh
```

```
  GNU nano 8.7.1              scripts/backup.sh *
#!/bin/bash
mkdir -p backup

# Jalankan pg_dump via docker
docker exec -t bab-5-postgres-db-1 pg_dump -U labuser labdb >>

# Buat file sha256 checksum
cd backup
sha256sum labdb.dump > labdb.dump.sha256
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ ./scripts/backup.sh
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ ls -lh backup/
total 8.0K
-rw-r--r-- 1 toimul toimul 2.6K Oct 10 15:16 labdb.dump
-rw-r--r-- 1 toimul toimul   77 Oct 10 15:16 labdb.dump.sha256
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ cd backup
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5/backup$ sha256sum --
check backup/*.dump.sha256
sha256sum: 'backup/*.dump.sha256': No such file or directory
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5/backup$ sha256sum --
check *.dump.sha256
labdb.dump: OK
```

Perintah G – Restore Test

Ada error

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ latest_dump="$(find
 backup -maxdepth 1 -name '*.dump' -type f | sort | tail -n 1)
"
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ ./scripts/restore-t
est.sh "$latest_dump"
sha256sum: labdb.dump: No such file or directory
labdb.dump: FAILED open or read
sha256sum: WARNING: 1 listed file could not be read
```

Perbaikan

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ sha256sum backup/la
bdb.dump > backup/labdb.dump.sha256
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ .scripts/restore-te
st.sh
-bash: .scripts/restore-test.sh: No such file or directory
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ ./scripts/restore-t
est.sh
Penggunaan: ./scripts/restore-test.sh backup/nama-file.dump
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db psql -U labuser -d labdb_restore_test -c "SELE
CT nrp, name FROM students ORDER BY nrp;"
psql: error: connection to server on socket "/var/run/postgres
ql/.s.PGSQL.5432" failed: FATAL:  database "labdb_restore_test
" does not exist
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db psql -U labuser -d postgres -c "CREATE DATABAS
E labdb_restore_test;"
CREATE DATABASE
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db psql -U labuser -d labdb_restore_test < backup
/labdb.dump
SET
SET
SET
SET
SET
 set_config
------------
```

```
 setval
-------
      2
(1 row)

ALTER TABLE
ALTER TABLE
CREATE INDEX
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
-T postgres-db psql -U labuser -d labdb_restore_test -c "SELE
CT nrp, name FROM students ORDER BY nrp;"
   nrp    |      name
----------+----------------
 31230001 | Mahasiswa Satu
 31230002 | Mahasiswa Dua
(2 rows)
```

Perintah H – Monitoring Dasar

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose ps
NAME                  IMAGE                     COMMAND
         SERVICE       CREATED          STATUS
    PORTS
bab-5-pgadmin-1       dpage/pgadmin4:latest     "/entrypoint.sh"
         pgadmin       34 minutes ago   Up 33 minutes
    443/tcp, 127.0.0.1:5050->80/tcp
bab-5-postgres-db-1   postgres:16-alpine        "docker-entrypoi
nt.s…"   postgres-db   34 minutes ago   Up 34 minutes (healthy
)   127.0.0.1:5432->5432/tcp
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose logs
 postgres-db --tail 100
postgres-db-1  |
postgres-db-1  | PostgreSQL Database directory appears to cont
ain a database; Skipping initialization
postgres-db-1  |
postgres-db-1  | 2026-10-10 08:04:31.498 UTC [1] LOG:  startin
g PostgreSQL 16.15 on x86_64-pc-linux-musl, compiled by gcc (A
lpine 15.2.0) 15.2.0, 64-bit
postgres-db-1  | 2026-10-10 08:04:31.499 UTC [1] LOG:  listeni
ng on IPv4 address "0.0.0.0", port 5432
postgres-db-1  | 2026-10-10 08:04:31.499 UTC [1] LOG:  listeni
ng on IPv6 address "::", port 5432
postgres-db-1  | 2026-10-10 08:04:31.509 UTC [1] LOG:  listeni
ng on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db pg_isready -U labuser -d labdb
/var/run/postgresql:5432 - accepting connections
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db psql -U labuser -d labdb -c "SELECT version();
"
                                              version

------------------------------------------------------------
----------------------------
 PostgreSQL 16.15 on x86_64-pc-linux-musl, compiled by gcc (Al
pine 15.2.0) 15.2.0, 64-bit
(1 row)
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker volume ls
DRIVER    VOLUME NAME
local     bab-3_pg-data
local     bab-5_pg-data
local     bab-5_pgadmin-data
local     data-vol
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker system df -v
Images space usage:

REPOSITORY       TAG          IMAGE ID       CREATED         S
IZE      SHARED SIZE   UNIQUE SIZE   CONTAINERS
httpd            2.4-alpine   b1ed1aaa73df   7 days ago      1
06MB     0B            106.3MB       0
bab-3-app        latest       6eaee8320cbd   2 weeks ago     2
20MB     2B            220MB         0
dpage/pgadmin4   latest       c332c5f6dfba   3 weeks ago     7
91MB     12.92MB       778MB         1
pens-web         1.0          71915eda7f49   4 weeks ago     7
2.2MB    2B            72.17MB       1
nginx            alpine       72ba65eb42c1   5 weeks ago     1
```

Perintah I – Cleanup Aman

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose down
[+] down 3/3
 ✔Container bab-5-pgadmin-1     Removed                  3.5s
 ✔Container bab-5-postgres-db-1 Removed                  0.4s
 ✔Network bab-5_data-net        Removed                  0.2s
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose up -
d postgres-db
[+] up 2/2
 ✔Network bab-5_data-net        Created                  0.0s
 ✔Container bab-5-postgres-db-1 Started                  0.3s
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db
requires at least 2 arg(s), only received 1
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db -U labuser --if-exists labdb_restore_test
OCI runtime exec failed: exec failed: unable to start containe
r process: exec: "-U": executable file not found in $PATH
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$ docker compose exec
 -T postgres-db dropdb -U labuser --if-exists labdb_restore_te
st
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-5$
```

Checklist Pass

[v] docker compose config tidak menghasilkan galat.

SS ke-2 Perintah B

[v] PostgreSQL berstatus healthy.

SS ke-1 Perintah C

[v] Tabel students dan data awal tersedia.

SS ke-1 dan ke-2 dari Perintah D

[v] pgAdmin dapat terhubung menggunakan hostname postgres-db.

SS ke-2 Script 5 & SS ke-2 Perintah C

[v] Data tetap tersedia setelah docker compose down dan up.

SS ke-2 Perintah E

[v] Backup dump berukuran lebih dari nol.

SS ke-3 Perintah F

[v] Checksum backup valid.

SS ke-4 Perintah F

[v] Restore berhasil pada database labdb_restore_test.

SS ke-1 Perintah G

[v] Jumlah dan isi record hasil restore sesuai database sumber.

SS ke-2 pada Perbaikan di Perintah G

[v] Credential dan dump tidak masuk repository Git.

SS ke-2 Script 2

Evaluasi Mandiri

1. Mengapa init script tidak dijalankan ulang saat volume lama masih ada?

   Docker image PostgreSQL dirancang hanya mengeksekusi script di /docker-entrypoint-initdb.d/ saat direktori data PostgreSQL masih kosong. Jika volume lama sudah ada, PostgreSQL melihat data atau database sudah terinisialisasi dan langsung melewatinya untuk mencegah penimpaan data.

2. Apa risiko menaruh password database pada docker-compose.yml?

   File docker-compose.yml umumnya di-commit ke dalam repositori Git. Jika password ditulis langsung di file tersebut, credential sensitif dapat terbaca oleh pihak lain. Praktikum ini mengatasinya dengan memisahkan kredensial ke file .env dan menyembunyikannya via .gitignore.

3. Bagaimana cara membuktikan backup dapat dipulihkan?

   Dengan menjalankan pengujian restore ke database pengujian terpisah (labdb_restore_test), lalu melakukan query untuk memastikan jumlah dan isi record pada database hasil restore identik dengan database sumber.

4. Apa bedanya logical backup pg_dump dan backup filesystem volume mentah?

   Logical Backup (pg_dump): Mengekstrak data menjadi perintah SQL atau skema. Berukuran lebih kecil, lebih fleksibel untuk di-restore ke versi PostgreSQL atau arsitektur berbeda, namun butuh waktu eksekusi saat pembuatan dan pemulihan.

   Filesystem Volume Backup: Mengopi langsung folder data fisik PostgreSQL . Pemulihannya sangat cepat, tetapi ukurannya besar, harus di-restore ke versi PostgreSQL yang sama persis, dan berisiko korupsi jika diambil saat database sedang aktif menulis.

5. Apa dampak docker compose down -v terhadap database?

Opsi -v (volumes) akan menghapus secara permanen named volume yang terhubung (bab-5_pg-data), sehingga seluruh data PostgreSQL di dalamnya akan hilang total. Jika ingin data tetap aman (persistent), gunakan perintah docker compose down tanpa opsi -v.